Measurements with highspeed ultrasound in different materials and with different applications

Nele Wurster and Hannes Winkler

Ultrasonic measurements in different materials were performed using an echoscope. The sound velocity in polyacryl was determined as $v_{1\mathrm{MHz}} = (2.619 \pm 0.032)\,\mathrm{km/s}$, $v_{2\mathrm{MHz}} = (2.677 \pm 0.033)\,\mathrm{km/s}$, and $v_{4\mathrm{MHz}} = (2.746 \pm 0.052)\,\mathrm{km/s}$, indicating dispersive behavior. In PVC, absorption coefficients of $\lambda_{1\mathrm{MHz}} = (0.0346 \pm 0.0313)\,\mathrm{cm}^{-1}$ and $\lambda_{2\mathrm{MHz}} = (0.0443 \pm 0.1297)\,\mathrm{cm}^{-1}$ were measured. Runtime measurements on a polyacrylic test block enabled the determination of internal structures with good agreement to caliper measurements, while higher ultrasonic frequencies showed improved spatial resolution. Using the Debye-Sears effect, the sound velocity in water was determined as $v_{\mathrm{water}} = (1438 \pm 65)\,\mathrm{m/s}$, in agreement with the literature value of $1440\,\mathrm{m/s}$. Overall, the experiments demonstrated the suitability of ultrasound as a non-destructive measurement technique.

Executed: 06^{th} May 2026
Submitted: 20^{th} May 2026

1 Introduction

Ultrasound describes the frequencies of sound waves, that are so large such that the human ear can not hear them. According it's inaudibility for the human, ultrasound was not utilized until late. Today ultrasound is used for spectroscopic examinations of materials and detections of objects, where other methods would be either to invasive (namely ultrasound in the medicin/materials science) or not precise enough (e.g. sonar).
In the following the propagation behavior of ultrasound in different materials will be examined, as well as measurements utilizing the properties of ultrasound.

2 Basic functions of an echoscope

An echoscope is an ultrasonic measurement system that connects a probe to a computer or an oscilloscope. The probe is commonly made of a transducer in form of a piezocristal. The echoscope then gives then an electical signal with a frequency to the probe which causes the piezocristal to vibrate and therefore it emits a supersonic wave. The probe not only emits ultrasound but it can act as a detector aswell when the piezocristal receives a vibration and converts it into an electical signal which then can be amplified and measured by the echoscope.
There are two main meathods of measurements using a echoscope: Firstly one probe emits a wave and another probe receives the incoming wave. Here the two probes have to be aligned perfectly to not get disturbances. Secondly one probe emits a wave and the same probe also receives it using the reflection at the end surface. The probe has to be aligned perpendicular to the end surface to not get disturbances.
There are multiple ways to process incoming signals from the echoscope. In this paper we want to highlight two. The most important one is the A-scan (amplitude scan) where the probe is kept at a constant position, measuring the amplitude of the emitted and received signal depending on the time after a pulse has been sent by the transmitter. Another way one can use an echoscope is the B-scan(brightness scan), where the probe is moved over an object and a 2D-Image of the plane in which teh probe was moved in is created. The amplitudes are given as different brightness values.

3 Experiment

3.1 Velocity of ultrasound in polyacryl

To measure the propagation speed of sound waves through media, three polyacryl cylinders with different lenghts were used. The time that the wave took to travel through the cylinder was measured via the method of reflection, where the ultrasound transmitter also worked as the detector of the signal. The time was then captured with the echoscope. With the method of reflection one has to notice that the length in the formula for the propagation speed is no double the length of the cylinder, concluding in:

$$v_{poly} = \frac{2 \cdot l}{\Delta t} \tag{1}$$

As noted above l is the total length of the cylinder, Δt is the time the wave took to travel twice through the cylinder and v_{poly} is the resulting speed of sound. This method was then applied to all three different cylinder-lengths, each with three different frequencies $f_{1MHz} = 1MHz$, $f_{2MHz} = 2MHz$ and $f_{4MHz} = 4MHz$. Since the absorption for f_{4MHz} was big, the largest cylinder could not be measured.
With the relevant data shown in 5.1, Tab. 4 the mean sound velocitys for each frequency is calculated:

$$v_{poly.1MHz} = (2.619 \pm 0.032)\frac{km}{s}$$
$$v_{poly.2MHz} = (2.677 \pm 0.033)\frac{km}{s}$$
$$v_{poly.4MHz} = (2.746 \pm 0.052)\frac{km}{s}$$

It is clearly visible, that the speed of sound in polyacryl depends on the frequency of the wave traveling through it. Thus it indicates dispersion in the sound velocity.
In the data sheet from a manufacutrer, the longitunal wave velocity for the frequency of 1Mhz is given by $v_{poly,Lit} = 2.7\frac{km}{s}$ [3], therefore the measured value does not match the literature within the error. This could be due to the fact that the measurement was done in saturation of the used piezoelectric ultrasound emitter. Also because of the absorption of the material, the gain on the echoscope had to be increased, which also increased the noise of the measurement thus increasing the uncertainty.

3.2 Absorption of ultrasound in PVC

The coefficient of absorption is a material-specific quantity which indicates how strongly a wave is attenuated propagading a medium. The attenuation follows an exponetial behavior. The intesity I as a function of the distance x can approximately be described by:

$$I(x) \sim e^{-\lambda x} \tag{2}$$

where λ is the absorption coefficient. To determine this coefficient, the signal's reflected amplitude was measured at different lengths of PVC cylinders. By fitting an exponential curve to the data, the absorption coefficient can be determined for the different frequencies.
For the $f = 1MHz$ messuarment, λ_{1MHz} was determined to be $\lambda_{1MHz} = (0.0346 \pm 0.0313)\frac{1}{cm}$ and for $f = 2MHz$ it was $\lambda_{2MHz} = (0.0443 \pm 0.1297)\frac{1}{cm}$.

The absorption coefficient increases with the frequency, which is a common behavior in materials. This can be explained by the fact that atoms oscillate faster at higher frequencies, leading to stronger scattering of the waves. The orderes wave increasingly transforms into disordered oscillations, resulting in energy loss.

3.3 Measurements on a polyacrylic test block

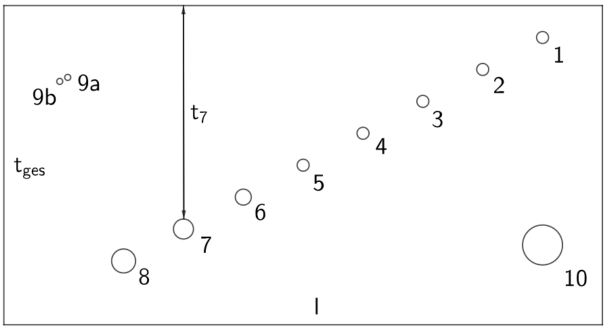

Figure 1: Schematics of the polyacrilic block

The polyacrylic block was investigated using the B-mode. The drilled holes are visible as bright circular regions, while the top and bottom surfaces of the block appear as horizontal lines.

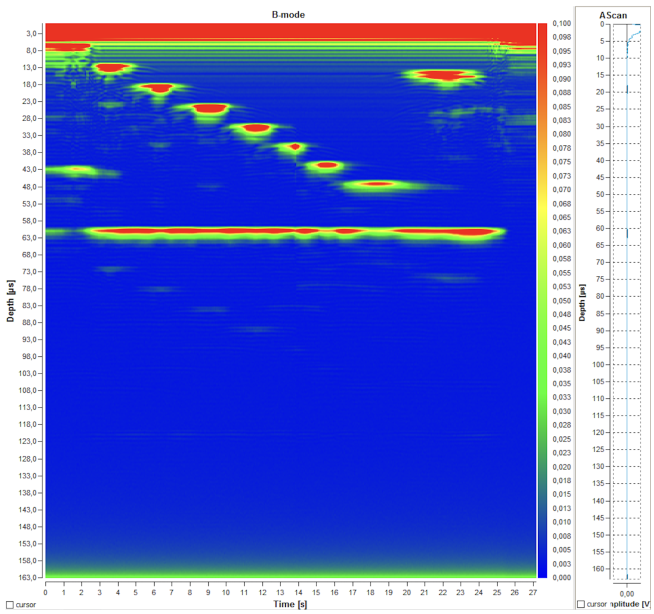

Figure 2: This figures show the B-mode of the polyacrylic block. The holes are visible as the red spots

As we know the speed of sound in polyacryl for the freuquency of $1MHz$ 3.1, we can use the echoscope to measure the depth d_i of the holes in the test block.

We used the $1MHz$ Probe for a clear runtime measurement. The depth can be calculated from the time difference Δt between the first and second reflection of the signal:

$$d_i = \frac{v_{poly,1MHz} \cdot \Delta t}{2} \quad (3)$$

For reference, the depth were mearured with a caliper. The results are shown in tab. 1.

i	$d_{i,caliper}$ [mm]	Δt [μs]	d_i [mm]
1	(7.00 ± 0.10)	(5.825 ± 0.010)	(7.63 ± 0.16)
2	(14.95 ± 0.10)	(11.606 ± 0.010)	(15.20 ± 0.23)
3	(22.95 ± 0.10)	(17.574 ± 0.010)	(23.01 ± 0.31)
4	(31.00 ± 0.10)	(23.358 ± 0.010)	(30.59 ± 0.40)
5	(39.05 ± 0.10)	(29.489 ± 0.010)	(38.62 ± 0.49)
6	(46.75 ± 0.10)	(35.182 ± 0.01)	(46.07 ± 0.58)
7	(54.15 ± 0.10)	(40.511 ± 0.010)	(53.05 ± 0.66)
8	(61.65 ± 0.10)	(46.058 ± 0.010)	(60.31 ± 0.75)
9a	(17.45 ± 0.10)	(14.567 ± 0.010)	(19.08 ± 0.27)
9b	(19.10 ± 0.10)	(15.906 ± 0.010)	(20.83 ± 0.29)
10	(55.45 ± 0.10)	(41.505 ± 0.010)	(54.35 ± 0.68)

Table 1: Measurement of the testblock

Most measured values agree within their margin of error. However, it is noticeable that holes 9a and 9b show deviating values. This is due to the fact that the holes are located very close to each other, meaning that a higher frequency would be beneficial in order to achieve a more precise resolution.

3.4 Resolution of an echoscope

As mentioned in 3.3, a higher frequency can be used to achieve a better resolution. To determine the resolution of the echoscope, the distance between two holes, ($9a$ und $9b$), was measured with different frequencies. The resolution can be defined as the minimum distance between two holes, where they can still be distinguished as two separate holes. Measurement for all three frequencies is shown in tab. 2.

Frequency	Distance [mm]
$2MHz$	(1.70 ± 0.13)
$4MHz$	(1.75 ± 0.14)

Table 2: Resolution of the echoscope

For a freuqueny of $1MHz$, the amplitude peaks could not be resolved as two distinct peaks; therefore, the distance between the reflectors could not be determined. In contrast, the measurements performed at the higher frequencies showed a clear separation of the two peaks, making the distance measurable and allowing the axial resolution to be determined from the spacing between the two holes. The resolution of the echoscope increases with increasing frequency, which is a characteristic behavior of ultrasound imaging systems. This can be explained by the shorter wavelengths associated with higher frequencies, enabling improved spatial resolution and therefore a better distinction between closely spaced objects. However, higher frequencies are also subject to stronger attenuation within the material, which reduces the penetration depth and consequently limits the effective measurement range in practical applications.

3.5 Spectral methods of an echoscope

To investigate the spectral methods, the reflection signal of a polyacrylic plate was analyzed using the A-scan. In addition, an FFT and the cepstrum of the signal were evaluated in order to determine the thickness of the plate. In the A-scan, several reflection peaks are visible, which originate from multiple reflections within the plate. After applying the FFT, approximately equidistant peaks appear in the frequency spectrum. From the spacing $\Delta\nu$ between neighboring peaks, the pulse travel time can be determined using

$$\Delta t = \frac{1}{\Delta \nu}$$

With the known speed of sound in the material, the thickness of the plate can then be calculated.

	$\Delta v[\frac{1}{\mu s}]$	$\Delta t[\mu s]$	$d[mm]$
runtime	–	(7.58 ± 0.01)	(10.15 ± 0.13)
FFT	(0.14 ± 0.10)	(7.14 ± 0.10)	(9.55 ± 0.12)
caliper	–	–	(10.00 ± 0.10)

Table 3: Speed of sound for different spectral methods

The values agree within the experimental uncertainties. However, no significant features could be observed in the cepstrum. This is likely caused by attenuation, noise, and weak higher-order reflections, which suppress the periodic structures required for a clear cepstral signal.

3.6 Debye-Sears effect

The Dabye-Sears effect describes an acousto-optic effect, where the ultrasound probe creates a standing wave with lattice constant $g = \lambda_S$ perpendicular to an incoming laser beam. The beam then diffracts as it does with a classical optical lattice. With the formula for the lattice spectrum from classical optics one can determine the wavelength λ_S of the ultrasound.

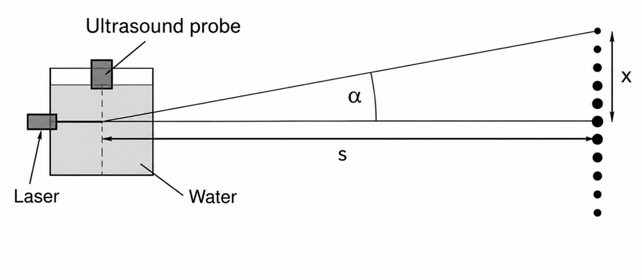

Figure 3: Setup used to measure the wavelength of ultrasound utilizing the Dabye-Sears effect

$$\lambda_S = \frac{n \cdot \lambda_{laser}}{sin(\alpha)} \overset{(1)}{\approx} \frac{n \cdot \lambda_{laser} \cdot s}{x} \tag{4}$$

In eq. 4 n is the number of the maxima, λ_{laser} is the wavelenght of the laser and α is the angle of diffraction. The approximation (1) in eq. 4 follows from the large difference in distance $x \ll s$.
The setup of the measurement is shown in fig. 3.
The measured maxima distance and the calculated sound velocities for every laser wavelenght can be found in 5.2, 5. Using the data from tab. 5, the mean speed of sound could be determined as $v_{S,water} = (1438 \pm 65)m/s$. Comparing this with the literature value of $v_{S,lit} = 1440m/s$ at a temperature of $T = 20°C$, one can see that the measurement was close to the literature which validates the measurement method.

4 Conclusion

The experiments investigated the propagation and application of ultrasound in different materials using an echoscope. The measurements of the sound velocity in polyacryl showed a dependence on frequency, indicating dispersive behavior of the material. In PVC, the absorption coefficient increased with frequency, confirming the expected stronger attenuation of high-frequency ultrasound. Ultrasonic runtime measurements on the polyacrylic test block enabled the determination of internal structures with good agreement to the caliper measurements. In addition, the resolution measurements demonstrated that higher frequencies improve the axial resolution due to shorter wavelengths, while simultaneously reducing the penetration depth because of increased attenuation. The FFT analysis also allowed the thickness of a polyacrylic plate to be determined successfully. Finally, the Debye-Sears experiment yielded a sound velocity in water consistent with the literature value. Overall, the experiments confirmed ultrasound as a reliable and versatile non-destructive method for material characterization and distance measurements.

References

[1] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Modulbeschreibungen/C-Modul/V35.pdf` last visited on 06.05.2026

[2] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Geraete/gs_200.pdf` last visited on 06.05.2026

[3] `https://www.3bscientific.com/product-manual/UE1070530_de.pdf` last visited on 14.05.2026

[4] `https://www.schweizer-fn.de/stoff/akustik/schallgeschwindigkeit.php` last visited on 18.05.2026

5 Appendix

5.1 Appendix A: Sound velocity in polyacryl

l [mm]	f [MHz]	Δt [μs]	v_{poly} [km/s]
41.30 ± 0.10	1	(32.0 ± 1.0)	(2.581 ± 0.081)
	2	(31.5 ± 1.0)	(2.622 ± 0.083)
	4	(30.0 ± 1.0)	(2.753 ± 0.092)
80.81 ± 0.10	1	(61.0 ± 1.0)	(2.650 ± 0.044)
	2	(60.5 ± 1.0)	(2.694 ± 0.045)
	4	(59.0 ± 1.0)	(2.739 ± 0.047)
120.76 ± 0.10	1	(92.0 ± 1.0)	(2.625 ± 0.029)
	2	(89.0 ± 1.0)	(2.714 ± 0.031)

Table 4: Data used for Experiment part 3.1, including lengths, frequencies, timedifferenze and corresponding sound velocity

5.2 Appendix B: Sound velocity in water

frequency [MHz]	sound velocity [km/s]		
	red laser: $\lambda_{laser,red} = 650nm$	green laser: $\lambda_{laser,green} = 532nm$	blue laser: $\lambda_{laser,blue} = 405nm$
3	2.22 ± 0.75	1.01 ± 0.34	1.15 ± 0.39
5	1.54 ± 0.31	1.26 ± 0.25	1.44 ± 0.29
7	1.62 ± 0.23	1.33 ± 0.19	1.47 ± 0.21
9	1.39 ± 0.15	1.52 ± 0.17	1.48 ± 0.17
11	1.46 ± 0.13	1.52 ± 0.14	1.41 ± 0.13
12	1.39 ± 0.12	1.40 ± 0.12	1.26 ± 0.11

Table 5: Sound velocities determined by the use of eq. 4 with given $lamda_{laser}$ and measured x for each frequency and laser wavelenght. n is 1 for all measurements.